Applied Econometrics: Session 5

Instrumental Variables, Non-Compliance, and LATE

PhD Intensive Course

Lecture 5

April 29, 2026

Why This Lecture Matters

Lectures 1–4 taught one recurring lesson: comparisons are causal only when the missing counterfactual is credible.

Today we study a different route to credibility:

Core idea

Use a variable Z_i that shifts treatment D_i, but is otherwise unrelated to potential outcomes.

- D_i may be selected on unobserved ability, preferences, income, or health.
- Z_i supplies cleaner variation in D_i.
- The price is that IV estimates a special effect: often the effect for compliers.

Roadmap for Today

1. Why IV?
2. Valid Instruments
3. Wald IV
4. 2SLS
5. Weak Instruments
6. Non-Compliance
7. LATE
8. Simulation
9. Review

The Problem: A Hidden Confounder

Suppose we want the effect of schooling on wages:

$$Y_i = \alpha + \beta D_i + u_i,$$

where Y_i is log wage and D_i is years of schooling.

- If ability is omitted, it is inside u_i.
- Ability likely affects both schooling and wages.
- Then $\mathrm{Cov}(D_i, u_i) \neq 0$, so OLS does not isolate β.

Translation

We are comparing people with different education, but also different unobserved traits.

Why More Controls May Not Save Us

The ideal long regression is

$$Y_i = \alpha + \beta D_i + W_i'\gamma + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid D_i, W_i] = 0.$$

But sometimes W_i is missing or badly measured.

- Ability, motivation, parental inputs, and networks are hard to observe.
- Proxy controls can reduce bias, but they rarely remove it mechanically.
- IV asks for a different object: a source of variation in D_i that is not chosen by the individual.

Practical lesson

IV is not a better regression control. It is a design argument about where treatment variation comes from.

The IV Picture

Think of three variables:

$$Z_i \longrightarrow D_i \longrightarrow Y_i.$$

A valid instrument has two roles:

1. It moves treatment: Z_i changes D_i.
2. It is clean: Z_i has no other route to Y_i.

A hidden confounder U_i may still affect both D_i and Y_i:

$$U_i \longrightarrow D_i, \qquad U_i \longrightarrow Y_i.$$

IV logic

We do not need D_i to be as good as random. We need the part of D_i caused by Z_i to be as good as random.

Examples of Instruments

- **Quarter of birth and schooling:** compulsory schooling laws make school-leaving age depend slightly on birth timing.
- **Draft lottery and military service:** random lottery number changes the probability of service.
- **Random offer and program take-up:** an encouragement changes who joins a program.
- **Distance to college and education:** proximity can reduce cost of schooling, but exclusion is controversial.

Student warning

"A variable that predicts treatment" is not enough. Many predictors of treatment are also predictors of potential outcomes.

Instrument or Control?

A control variable is something we condition on.

$$Y_i = \alpha + \beta D_i + X_i'\gamma + e_i.$$

An instrument is something we use to generate treatment variation.

$$D_i = \pi_0 + \pi_1 Z_i + X_i'\pi_2 + v_i.$$

- Controls block back-door paths from D_i to Y_i.
- Instruments create front-door-like variation from Z_i to D_i.
- You can use controls inside an IV design, but the identifying variation still comes from Z_i.

Notation for the Basic IV Model

Start with

$$Y_i = \alpha + \beta D_i + u_i, \qquad \mathrm{Cov}(D_i, u_i) \neq 0.$$

We observe an instrument Z_i.

Three core conditions

1. **Relevance:** $\mathrm{Cov}(Z_i, D_i) \neq 0$.
2. **Independence:** Z_i is as good as randomly assigned relative to potential outcomes and unobservables.
3. **Exclusion:** Z_i affects Y_i only through D_i.

Appendix details

Condition 1: Relevance

Relevance means the instrument changes treatment:

$$\mathrm{Cov}(Z_i, D_i) \neq 0.$$

Estimate the first stage:

$$D_i = \pi_0 + \pi_1 Z_i + r_i.$$

- If $\pi_1 = 0$, the instrument gives no information about treatment.
- If π_1 is tiny, IV estimates are unstable and noisy.
- Relevance can be checked in the data.

Classroom phrase

The first stage asks: does the instrument actually move the treatment?

Condition 2: Independence

Independence means Z_i is not chosen because of the same unobserved forces that determine Y_i.
A strong version is

$$Z_i \perp\!\!\!\perp \{Y_i(d), D_i(z) : d, z \in \{0, 1\}\}.$$

- Random assignment of encouragement often makes this plausible.
- Institutional rules can make this plausible, but not automatic.
- Balance checks help detect problems, but cannot prove independence.

Interpretation

Independence is about who gets the instrument, not who gets the treatment.

Condition 3: Exclusion Restriction

Exclusion says the instrument has no direct effect on the outcome.

$$Z_i \to D_i \to Y_i, \qquad \text{but not} \qquad Z_i \to Y_i.$$

In potential-outcome notation:

$$Y_i(d, z) = Y_i(d) \quad \text{for all } d, z.$$

- A schooling instrument must not affect wages except through schooling.
- A program encouragement must not affect outcomes through information, attention, or stigma except through participation.
- Exclusion is usually argued from institutional knowledge, not tested directly.

What Can Be Tested?

Requirement	Data check?	What to do in practice
Relevance	Yes	Report first-stage coefficient and F-statistic
Independence	Partly	Check balance on pre-treatment covariates
Exclusion	No direct test	Defend the institutional story; discuss threats
Monotonicity	Usually no direct test	Argue no one moves opposite to the instrument

Exam point

A significant first stage does not validate the instrument. It validates only one required condition.

Bad Instruments: Common Failure Modes

- **Weak instrument:** Z_i barely moves D_i.
- **Direct effect:** Z_i affects Y_i through a path other than treatment.
- **Selection into instrument:** people with different potential outcomes are more likely to receive Z_i.
- **Multiple treatments:** Z_i changes several behaviors, not just the named treatment.

Practical rule

Always write the story as a causal graph in words. If you cannot explain the path from Z_i to D_i and rule out other paths to Y_i, do not trust IV.

The First Stage

The first stage estimates how treatment changes with the instrument:

$$D_i = \pi_0 + \pi_1 Z_i + r_i.$$

For binary Z_i,

$$\pi_1 = \mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0].$$

- If D_i is schooling, the first stage is extra years of schooling caused by Z_i.
- If D_i is program take-up, the first stage is the compliance rate induced by assignment.

Units

First stage has units of treatment per unit of instrument.

The Reduced Form

The reduced form estimates how the outcome changes with the instrument:

$$Y_i = \rho_0 + \rho_1 Z_i + \eta_i.$$

For binary Z_i,

$$\rho_1 = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0].$$

- The reduced form is an effect of the instrument on the outcome.
- Under exclusion, that effect must operate through treatment.
- It is often called an intention-to-treat effect in encouragement designs.

Units

Reduced form has units of outcome per unit of instrument.

The Wald Estimator

If Z_i is binary, the IV estimator is the ratio:

$$\hat{\beta}_{Wald} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}.$$

Equivalently,

$$\beta_{IV} = \frac{\mathrm{Cov}(Y_i, Z_i)}{\mathrm{Cov}(D_i, Z_i)} = \frac{\text{reduced form}}{\text{first stage}}.$$

Intuition

Convert the effect of the instrument on the outcome into outcome units per one unit of treatment.

Appendix details

A Numerical Wald Example

Suppose assignment to a tutoring offer changes tutoring take-up and exam scores:

	Z=1	Z=0
$\mathbb{E}[D \mid Z]$	0.70	0.30
$\mathbb{E}[Y \mid Z]$	74	70

First stage:

$$0.70 - 0.30 = 0.40.$$

Reduced form:

$$74 - 70 = 4.$$

Wald IV:

$$\frac{4}{0.40} = 10.$$

Interpretation: among the students whose tutoring status was changed by the offer, tutoring raised scores by 10 points.

Why the Ratio Makes Sense

Imagine the offer raises average test scores by 4 points.

- But only 40 percentage points of students change tutoring status because of the offer.
- The observed 4-point difference is diluted by students whose treatment did not change.
- Dividing by 0.40 undilutes the assignment effect into a treatment effect.

Careful language

This is not the effect for everyone unless treatment effects are constant or the complier effect equals the population effect.

IV with Controls

In many applications we include controls X_i:

$$Y_i = \alpha + \beta D_i + X_i'\gamma + u_i.$$

The instrument must be valid conditional on those controls.

- First stage: regress D_i on Z_i and X_i.
- Reduced form: regress Y_i on Z_i and X_i.
- IV ratio uses the residual variation in Z_i after removing X_i.

Rule

Any exogenous control in the outcome equation belongs in the first stage too.

Appendix details

Two-Stage Least Squares: Stage 1

Stage 1 predicts treatment using the instrument and controls:

$$D_i = \pi_0 + \pi_1 Z_i + X_i' \pi_2 + r_i.$$

Get fitted values:

$$\hat{D}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i + X_i' \hat{\pi}_2.$$

- $\hat{D}_i$ is the part of treatment explained by Z_i and X_i.
- The key identifying variation in $\hat{D}_i$ comes from Z_i.

Two-Stage Least Squares: Stage 2

Stage 2 regresses the outcome on predicted treatment and controls:

$$Y_i = \alpha + \beta \hat{D}_i + X_i' \gamma + \text{error}_i.$$

- The coefficient on $\hat{D}_i$ is the 2SLS estimate.
- With one endogenous variable and one instrument, it equals the Wald ratio after residualizing controls.
- Software must compute the standard errors; manual second-stage OLS standard errors are wrong.

Appendix details

What 2SLS Is Doing Geometrically

OLS uses all variation in D_i:

$$D_i = \text{clean variation} + \text{confounded variation}.$$

2SLS keeps the variation in D_i that is explained by the instrument:

$$\hat{D}_i = \mathrm{Projection}(D_i \mid Z_i, X_i).$$

Intuition

If Z_i is clean, then the instrument-generated part of treatment is clean.

- 2SLS does not make every treated and untreated person comparable.
- It compares people whose treatment differs because of Z_i.

Multiple Instruments

If there are several instruments $Z_i = (Z_{1i}, Z_{2i}, \ldots, Z_{ki})$, stage 1 is

$$D_i = \pi_0 + Z_i'\pi_1 + X_i'\pi_2 + r_i.$$

- Multiple instruments can strengthen prediction of treatment.
- They also multiply the number of exclusion restrictions to defend.
- With too many weak or questionable instruments, 2SLS can move toward the biased OLS estimate.

Practical rule

More instruments are not automatically better. Each instrument needs its own validity argument.

Do Not Run IV by Hand for Inference

Manually running the two regressions is useful for intuition:

$$D_i \to \hat{D}_i, \qquad Y_i \text{ on } \hat{D}_i.$$

But the second-stage residuals are not ordinary OLS residuals.

- The second stage uses an estimated regressor.
- Naive OLS standard errors ignore first-stage uncertainty.
- Use an IV/2SLS routine for final tables.

For this course

By hand for understanding; proper IV software for reported inference.

Weak Instruments

A weak instrument has a small first stage:

$$D_i = \pi_0 + \pi_1 Z_i + r_i, \qquad \pi_1 \approx 0.$$

- The denominator of the IV ratio is small.
- Small sampling noise in the reduced form gets magnified.
- Confidence intervals become wide and conventional approximations can be misleading.

Visual intuition

Weak IV is like estimating a slope by dividing by a nearly flat first-stage line.

Appendix details

The First-Stage F-Statistic

For one endogenous treatment, a common diagnostic is the first-stage F-statistic on the excluded instruments.

- Very small F: the instrument barely shifts treatment.
- Larger F: more information about treatment from the instrument.
- A classic rule of thumb is $F > 10$, but it is not a theorem of validity.

Important distinction

Strong first stage helps precision and weak-IV bias. It does not prove exclusion or independence.

Weak-IV Symptoms

In empirical work, weak instruments often produce:

- large standard errors,
- estimates that change a lot across specifications,
- implausibly large point estimates from tiny first stages,
- IV estimates drifting toward OLS in finite samples,
- confidence intervals that are not trustworthy under standard approximations.

Practical response

Report the first stage clearly, show robustness, and be honest if the design has little treatment variation.

What IV Does Not Fix

IV does not solve every causal problem.

- It does not fix a bad exclusion restriction.
- It does not identify effects for people whose treatment is unaffected by the instrument.
- It does not make finite samples behave like large samples.
- It does not remove the need for a clear research question.

Bottom line

IV trades one impossible requirement, observing all confounders, for another demanding requirement, finding valid instrument variation.

Assignment Versus Treatment Received

Many experiments randomize an assignment Z_i, but people do not perfectly follow it.

$$Z_i = \text{assigned treatment}, \qquad D_i = \text{received treatment}.$$

- Some assigned to treatment do not take it.
- Some assigned to control may obtain treatment anyway.
- Randomization applies to Z_i, not necessarily to D_i.

Key shift

With non-compliance, random assignment becomes an instrument for treatment received.

Why Not Just Compare by Treatment Received?

Comparing Y_i by received treatment D_i can reintroduce selection:

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0].$$

- People who comply may be richer, healthier, more motivated, or more informed.
- People who refuse treatment may differ in unobserved baseline outcomes.
- Treatment received is partly chosen, even if assignment was random.

Example

If only users with new phones can receive a push notification, comparing delivered versus not delivered also compares richer versus poorer users.

Intention-to-Treat

Comparing by assignment is still causal:

$$\mathrm{ITT}_Y = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0].$$

- This is the causal effect of being assigned or encouraged.
- It is often policy-relevant: what happens if we send offers?
- But it is diluted when many assigned people do not take treatment.

Question

If we want the effect of actually receiving treatment, how do we undo the dilution?

Compliance Types

Define potential treatment received:

$$D_i(1) = \text{treatment if assigned}, \qquad D_i(0) = \text{treatment if not assigned}.$$

Type	$D_i(1)$	$D_i(0)$	Meaning
Complier	1	0	follows assignment
Never-taker	0	0	never takes treatment
Always-taker	1	1	always takes treatment
Defier	0	1	does the opposite of assignment

Which Types Move the First Stage?

The first-stage effect of assignment on treatment is

$$\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0].$$

Under random assignment, this equals

$$\mathbb{E}[D_i(1) - D_i(0)].$$

- Compliers contribute $1 - 0 = 1$.
- Never-takers contribute $0 - 0 = 0$.
- Always-takers contribute $1 - 1 = 0$.
- Defiers contribute $0 - 1 = -1$.

Monotonicity

Monotonicity rules out defiers:

$$D_i(1) \geq D_i(0) \quad \text{for every } i.$$

- Assignment can encourage treatment, or do nothing.
- It cannot make someone less likely to take treatment.
- This is often plausible for offers, eligibility, reminders, or draft risk.

Why it matters

Without monotonicity, positive and negative first-stage responses can cancel, and the IV ratio is hard to interpret.

Local Average Treatment Effect

Under independence, exclusion, relevance, and monotonicity:

$$\frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)].$$

Name

This is the Local Average Treatment Effect, or LATE: the average treatment effect for compliers.

Appendix details

Why IV Identifies Compliers

The instrument only reveals treatment effects for people whose treatment status changes when Z_i changes.

- Never-takers are untreated regardless of assignment.
- Always-takers are treated regardless of assignment.
- Compliers switch treatment status when assignment changes.

Core intuition

If the instrument does not change your treatment, your treatment effect cannot show up in the IV numerator.

LATE Is Not Automatically ATE

The ATE is

$$\mathrm{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)].$$

LATE is

$$\mathrm{LATE} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}].$$

- If treatment effects are constant, LATE equals ATE.
- If treatment effects differ by compliance type, LATE may be far from ATE.
- The IV estimate is internally valid for the complier group identified by that instrument.

External validity question

Are the compliers the people policy makers care about?

Interpreting Compliers

Compliers are defined relative to the instrument.

- With a tuition subsidy, compliers are students induced to enroll by the subsidy.
- With quarter of birth, compliers are people whose schooling changes because school-entry and leaving rules bind.
- With a push notification assignment, compliers are users who receive the push only when assigned.

Important

Different instruments for the same treatment can estimate different LATEs.

A Non-Compliance Numeric Example

Suppose an online course randomly offers free tutoring.

	Z=1	Z=0
$\mathbb{E}[D \mid Z]$	0.60	0.10
$\mathbb{E}[Y \mid Z]$	82	77

$$\mathrm{ITT}_D = 0.60 - 0.10 = 0.50, \qquad \mathrm{ITT}_Y = 82 - 77 = 5.$$

$$\widehat{\mathrm{LATE}} = \frac{\mathrm{ITT}_Y}{\mathrm{ITT}_D} = \frac{5}{0.50} = 10.$$

The offer raises scores by 5 points; tutoring raises scores by 10 points for students induced into tutoring by the offer.

What To Report in an IV Paper

A credible IV table should not show only the second-stage coefficient.

- State the instrument and the treatment it shifts.
- Report first-stage coefficient and F-statistic.
- Report reduced form when useful.
- Explain independence and exclusion in institutional detail.
- State the likely complier population.
- Discuss why LATE is or is not the policy-relevant estimand.

Python Example for This Lecture

The script for this lecture is:

`lectures/code/lecture-5-iv-late.py`.

It simulates four lessons:

1. OLS is biased when treatment is correlated with an omitted variable.
2. IV recovers the treatment effect when the instrument is valid and strong.
3. Weak first stages make IV estimates unstable.
4. With non-compliance, Wald IV recovers LATE for compliers.

Why simulation helps

We know the true data-generating process, so students can see which estimator fails and why.

Simulation Design: Endogeneity and IV

The script generates

$$U_i \sim N(0,1), \qquad Z_i \sim \text{Bernoulli}(1/2),$$

$$D_i = 0.8Z_i + 0.7U_i + v_i,$$

$$Y_i = 2D_i + 1.5U_i + e_i.$$

- True treatment effect is 2.
- OLS is biased because D_i contains U_i.
- IV uses only the variation in D_i caused by Z_i.

Simulation Design: Non-Compliance

For LATE, the script creates compliance types:

$$(D_i(1), D_i(0)) \in \{(1,0), (0,0), (1,1)\}.$$

- Compliers follow assignment.
- Never-takers never receive treatment.
- Always-takers receive treatment regardless of assignment.
- No defiers are generated, so monotonicity holds.

Treatment effects are heterogeneous, so the complier effect differs from the population ATE.

What Students Must Remember

1. IV is for endogeneity from unobserved confounding, reverse causality, or measurement error.
2. A valid instrument must be relevant, independent, and excluded from the outcome equation.
3. Wald IV is reduced form divided by first stage.
4. 2SLS operationalizes the same idea with controls and multiple instruments.
5. Weak instruments are dangerous even when exclusion is conceptually plausible.
6. With non-compliance and monotonicity, IV estimates LATE for compliers.

Common Exam Mistakes

- Saying "Z is correlated with Y, so it is a good instrument".
- Forgetting that Z must move D, not just predict Y.
- Treating exclusion as a statistical test.
- Interpreting LATE as the ATE for everyone.
- Ignoring the first stage.
- Using treatment received as if it were randomized in a non-compliance setting.

Source and Further Reading

This lecture is based on ideas from:

- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 08, "Instrumental Variables".
- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 09, "Non Compliance and LATE".
- Angrist and Pischke, *Mostly Harmless Econometrics*.
- Angrist and Pischke, *Mastering 'Metrics*.

Handbook URLs:

- `https://matheusfacure.github.io/python-causality-handbook/08-Instrumental-Variables.html`
- `https://matheusfacure.github.io/python-causality-handbook/09-Non-Compliance-and-LATE.html`

Appendix Map

- A1: IV moment condition from the structural equation
- A2: Wald estimator derivation
- A3: 2SLS projection formula
- A4: IV with controls by residualizing
- A5: Why weak instruments inflate variance
- A6: LATE derivation under monotonicity
- A7: Compliance shares from first-stage moments

Appendix A1: IV Moment Condition

Start from the structural equation

$$Y_i = \alpha + \beta D_i + u_i.$$

OLS fails when

$$\mathbb{E}[D_i u_i] \neq 0.$$

A valid instrument satisfies

$$\mathbb{E}[Z_i u_i] = 0, \qquad \mathbb{E}[Z_i D_i] \neq 0$$

after demeaning or after including an intercept.
Multiply the structural equation by Z_i and take expectations:

$$\mathbb{E}[Z_i Y_i] = \alpha \mathbb{E}[Z_i] + \beta \mathbb{E}[Z_i D_i] + \mathbb{E}[Z_i u_i].$$

With de-meaned variables, $\mathbb{E}[Z_i] = 0$ and $\mathbb{E}[Z_i u_i] = 0$, so

$$\beta = \frac{\mathbb{E}[Z_i Y_i]}{\mathbb{E}[Z_i D_i]}.$$

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Appendix A2: Wald Estimator

For a binary instrument,

$$\mathrm{Cov}(Y_i, Z_i) = \mathbb{P}(Z_i = 1)\mathbb{P}(Z_i = 0)\left(\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]\right).$$

Similarly,

$$\mathrm{Cov}(D_i, Z_i) = \mathbb{P}(Z_i = 1)\mathbb{P}(Z_i = 0)\left(\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]\right).$$

Therefore the probability factor cancels:

$$\frac{\mathrm{Cov}(Y_i, Z_i)}{\mathrm{Cov}(D_i, Z_i)} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}.$$

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Appendix A3: 2SLS Projection Formula

Let X include exogenous controls and the intercept, D be the endogenous treatment, and Z include excluded instruments plus X.
The first-stage projection matrix is

$$P_Z = Z(Z'Z)^{-1}Z'.$$

The fitted treatment is

$$\hat{D} = P_Z D.$$

The 2SLS coefficient on D can be written as

$$\hat{\beta}_{2SLS} = (D'P_Z M_X D)^{-1} D'P_Z M_X Y,$$

where M_X removes the included exogenous controls.
In the just-identified no-control case, this collapses to

$$\hat{\beta}_{2SLS} = \frac{Z'Y}{Z'D},$$

after demeaning.

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Appendix A4: IV with Controls

Suppose the model is

$$Y_i = \alpha + \beta D_i + X_i'\gamma + u_i.$$

Let $\tilde{Y}_i$, $\tilde{D}_i$, and $\tilde{Z}_i$ be residuals from separate regressions of Y_i, D_i, and Z_i on X_i. By Frisch-Waugh-Lovell, the controlled IV estimand is

$$\beta_{IV|X} = \frac{\mathrm{Cov}(\tilde{Z}_i, \tilde{Y}_i)}{\mathrm{Cov}(\tilde{Z}_i, \tilde{D}_i)}.$$

- The numerator is the reduced form after removing X_i.
- The denominator is the first stage after removing X_i.
- The instrument must be valid conditional on X_i.

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Appendix A5: Weak-IV Variance Intuition

In the just-identified de-meaned case,

$$\hat{\beta}_{IV} = \frac{\widehat{\mathrm{Cov}}(Z_i, Y_i)}{\widehat{\mathrm{Cov}}(Z_i, D_i)}.$$

Let

$$\pi = \frac{\mathrm{Cov}(Z_i, D_i)}{\mathrm{Var}(Z_i)}$$

be the first-stage slope. When π is close to zero, the denominator is close to zero.
A rough delta-method approximation gives

$$\mathrm{Var}(\hat{\beta}_{IV}) \propto \frac{1}{\pi^2}.$$

Consequence

Halving the first-stage strength can roughly quadruple the variance, before considering finite-sample bias.

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Appendix A6: LATE Derivation

Let $D_i(z)$ be potential treatment under assignment z, and $Y_i(d)$ potential outcome under treatment d.
Observed treatment:

$$D_i = Z_i D_i(1) + (1 - Z_i) D_i(0).$$

By exclusion, observed outcome is

$$Y_i = Y_i(0) + D_i\{Y_i(1) - Y_i(0)\}.$$

By independence,

$$\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0] = \mathbb{E}\left[(D_i(1) - D_i(0))(Y_i(1) - Y_i(0))\right].$$

With monotonicity, $D_i(1) - D_i(0) \in \{0, 1\}$, so the numerator equals

$$\mathbb{P}(\text{complier})\mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}].$$

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Appendix A6 Continued: Denominator

The first stage is

$$\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0] = \mathbb{E}[D_i(1) - D_i(0)].$$

Under monotonicity,

$$\mathbb{E}[D_i(1) - D_i(0)] = \mathbb{P}(\text{complier}).$$

Therefore

$$\frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}].$$

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Appendix A7: Compliance Shares

Under random assignment:

$$\mathbb{E}[D_i \mid Z_i = 1] = \mathbb{E}[D_i(1)], \qquad \mathbb{E}[D_i \mid Z_i = 0] = \mathbb{E}[D_i(0)].$$

If monotonicity holds:

$$\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0] = \mathbb{P}(\text{complier}).$$

If there are no always-takers, then

$$\mathbb{E}[D_i \mid Z_i = 0] = 0, \qquad \mathbb{E}[D_i \mid Z_i = 1] = \mathbb{P}(\text{complier}).$$

If always-takers exist, $\mathbb{E}[D_i \mid Z_i = 0]$ estimates their share.

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